Inventory Management System

Pre-requisites:

- ✔ Git installed and configured with SSH or Personal Access Token (PAT)
- ✔ Python installed (preferably 3.x)
- ✔ Azure DevOps account with a new project created
- ✔ Your **local project folder** contains:
  - check_reorder.py
  - final_inventory_dashboard.csv
  - azure-pipelines.yml

Step 1: Create Python and YAML Files

1. Python Script (check_reorder.py)

```
import pandas as pd

# Load the inventory CSV file
df = pd.read_csv("final_inventory_dashboard.csv")

# Filter only products that need reorder
reorder_df = df[df["reorder_flag"] == "REORDER"]

# Save the result to a new CSV file
reorder_df.to_csv("reorder_report.csv", index=False)

print("reorder_report.csv generated successfully.")
```

⚙ 2. Azure DevOps Pipeline File (azure-pipelines.yml)

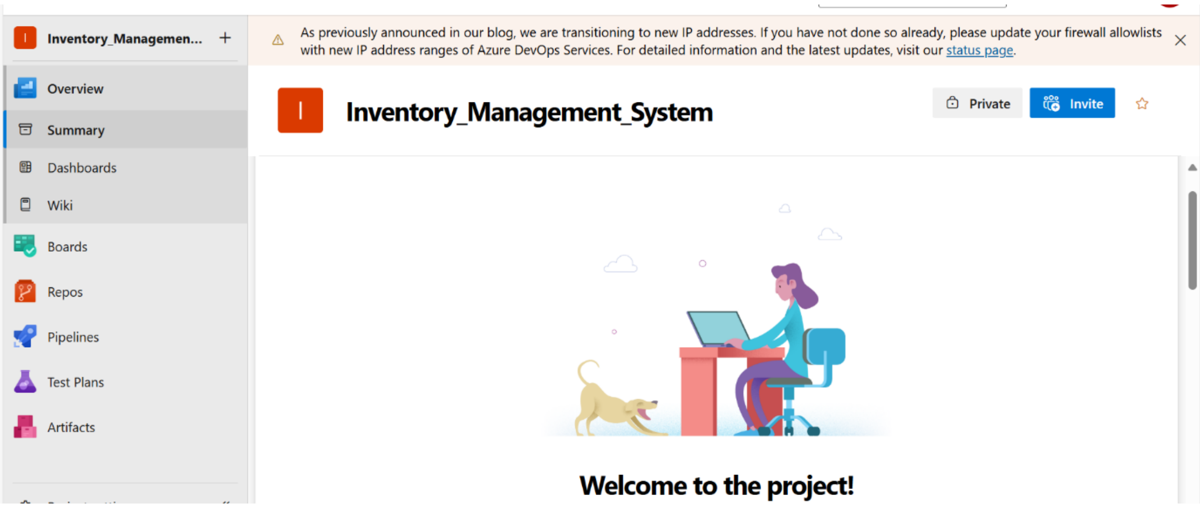
```
trigger:
  schedule:
    - cron: "0 7 * * *"  # Runs daily at 7 AM UTC
      displayName: Daily 7AM run
      branches:
        include:
          - main
```

```yaml
    always: true

pool:
  vmImage: 'ubuntu-latest'

steps:
  - task: UsePythonVersion@0
    inputs:
      versionSpec: '3.10'
      addToPath: true

  - script: |
      python -m pip install --upgrade pip
      pip install pandas
    displayName: 'Install Dependencies'

  - script: |
      python check_reorder.py
    displayName: 'Run Reorder Check Script'

  - task: PublishBuildArtifacts@1
    inputs:
      PathtoPublish: 'reorder_report.csv'
      ArtifactName: 'ReorderReport'
      publishLocation: 'Container'
    displayName: 'Publish Reorder Report Artifact'
```

Step 2: Create a New Azure DevOps Project

1. Go to https://dev.azure.com/
2. Click **"New Project"**

3. Give it a name like Inventory_Management_System
4. Set Visibility: Private
5. Click **Create**



Step 3: Push Local Project to Azure Repo via SSH

In PowerShell:

cd "C:\Users\anith\OneDrive\Desktop\Inventory_Management_System\Inventory_Management_System-2"

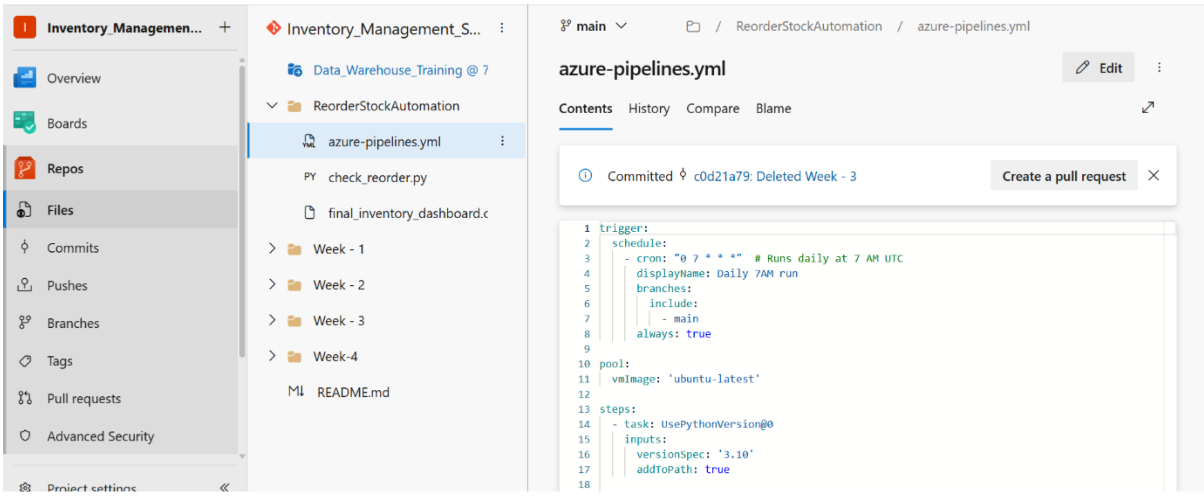
git init

git remote add origin git@ssh.dev.azure.com:v3/rithi31/Inventory_Management_System/Inventory_Management_System

git add .

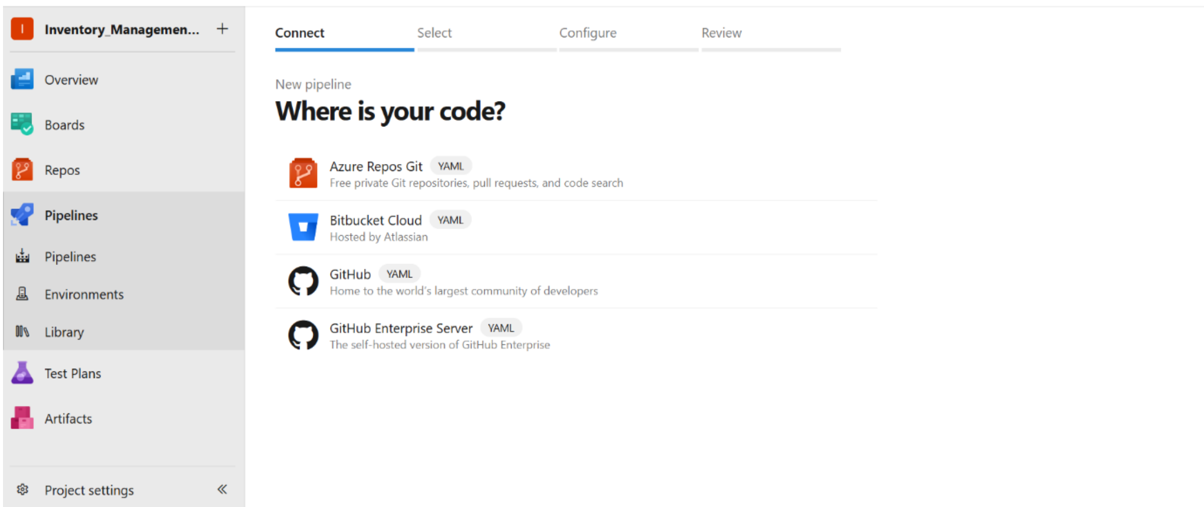
git commit -m "Initial commit"

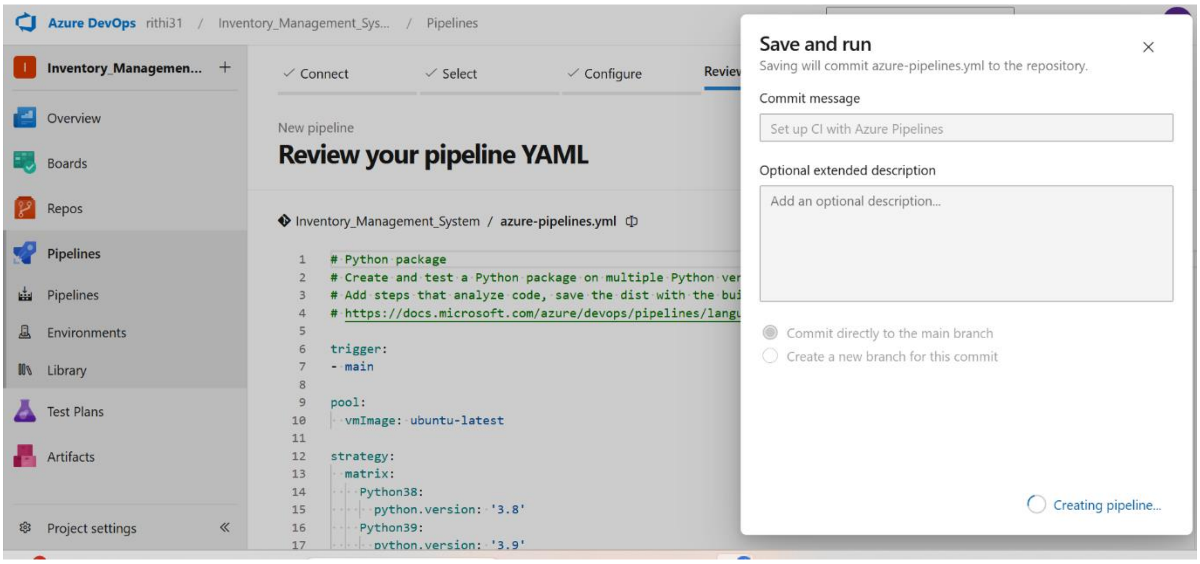
git push -u origin main



Step 4: Configure and Run Azure Pipeline

1. Go to your **Azure DevOps project**
2. Click **Pipelines → Create Pipeline**
3. Choose:
   - **Azure Repos Git**
   - Select your repository
   - **"Existing Azure Pipelines YAML file"**
4. Choose:
   - Branch: main
   - Path: /azure-pipelines.yml
5. Click **Continue**, then **Run Pipeline**





Step 5: Final Output

- The pipeline will:
  - Install dependencies
  - Run the reorder script
  - Save reorder_report.csv